

CS & IT ENGINEERING



Computer Network

Routing

Lecture No. - 04

By - Abhishek Sir





Recap of Previous Lecture



Topic

Distance Vector Routing





Topics to be Covered



Topic

Distance Vector Routing



ABOUT ME



Hello, I'm **Abhishek**

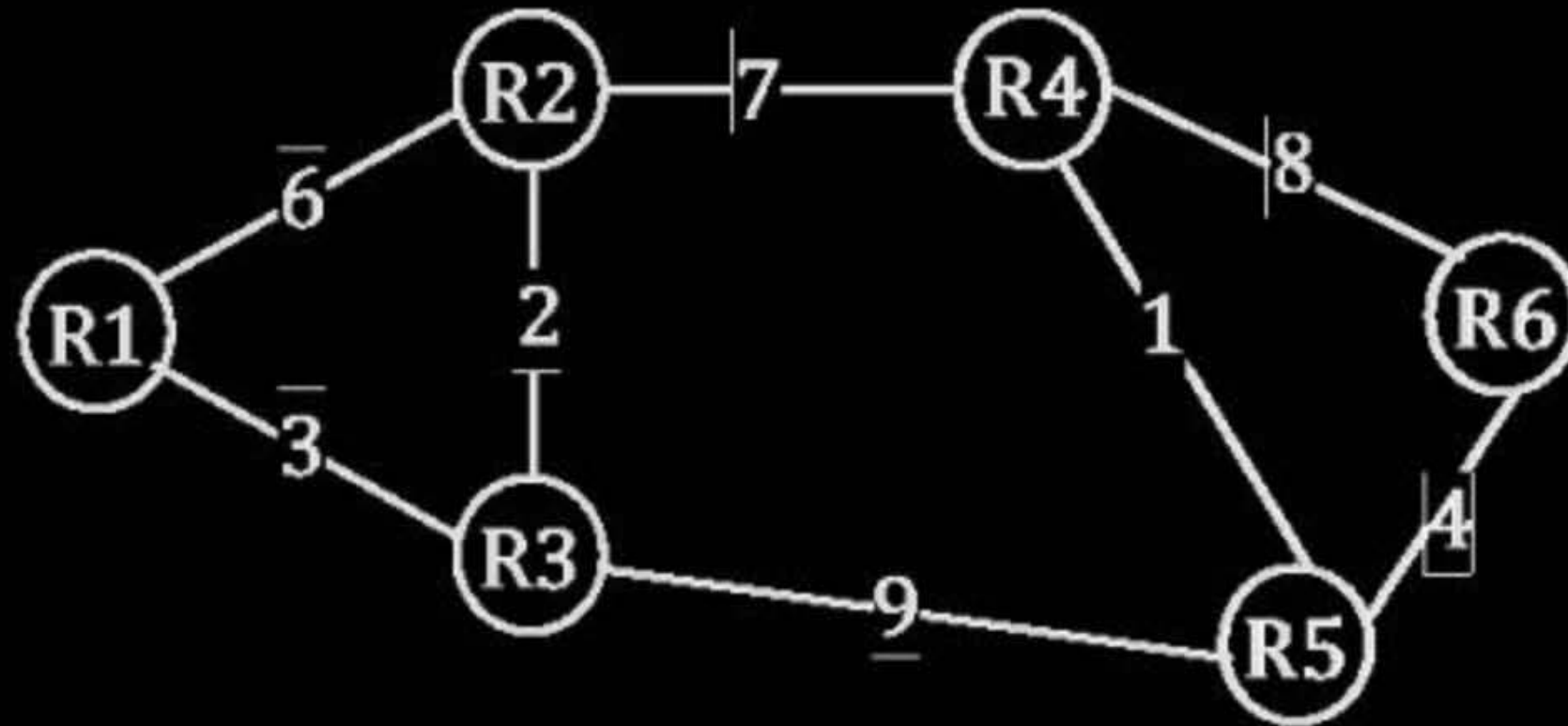
- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW



Statement for linked question

Consider a network with 6 routers R1 to R6 connected with links having weights as shown in the following diagram:



#Q. All the routers use the distance vector based routing algorithm to update their routing tables. Each router starts with its routing table initialized to contain an entry for each neighbor with the weight of the respective connecting link. After all the routing tables stabilize, how many links in the network will never be used for carrying any data?

[GATE-2010]

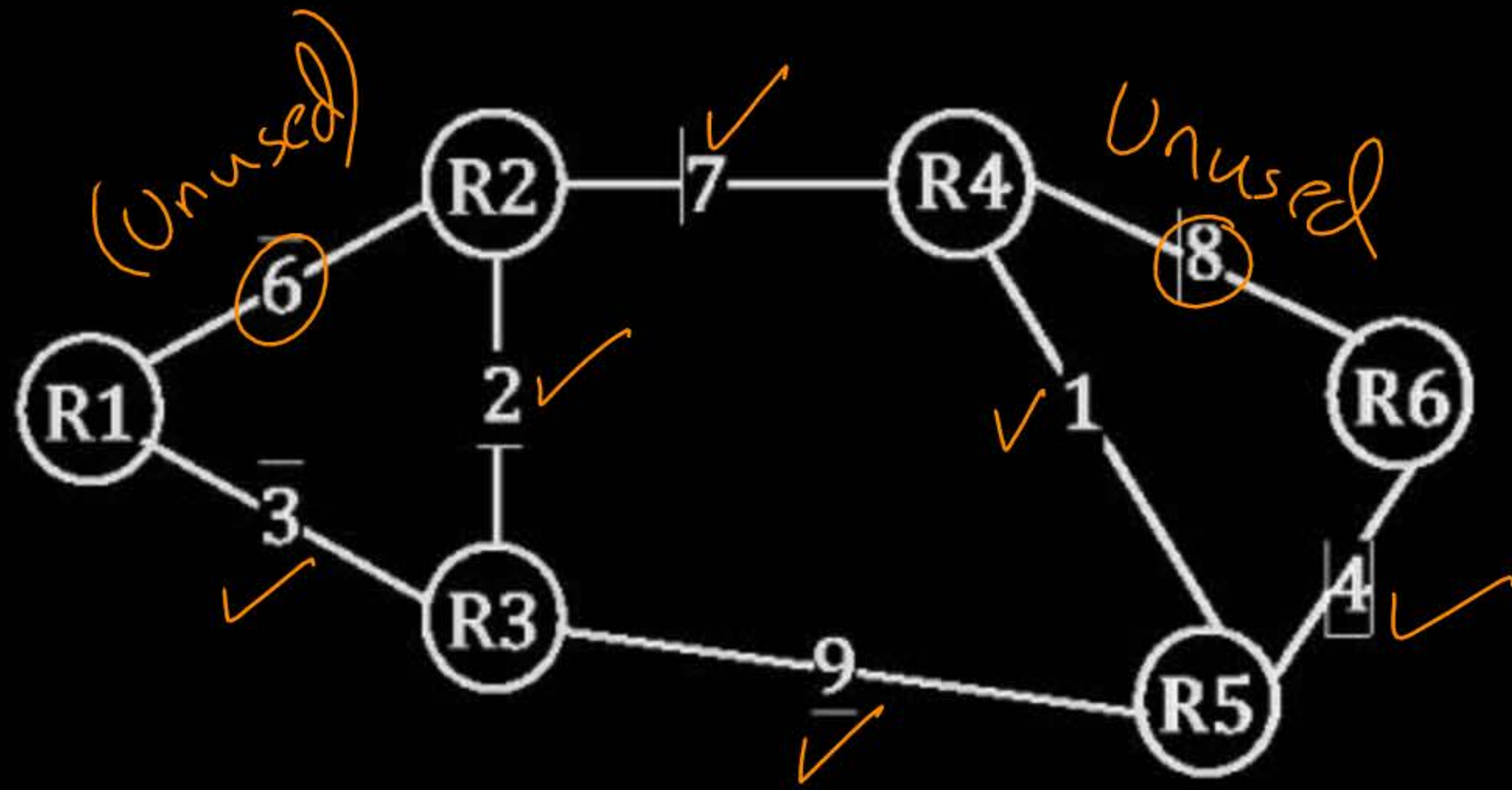
(A) 4

(B) 3

☒ (C) 2

(D) 1

Ans: C



#Q. Suppose the weights of all unused links in the previous question are changed to 2 and the distance vector algorithm is used again until all routing tables stabilize. How many links will now remain unused?

[GATE-2010]

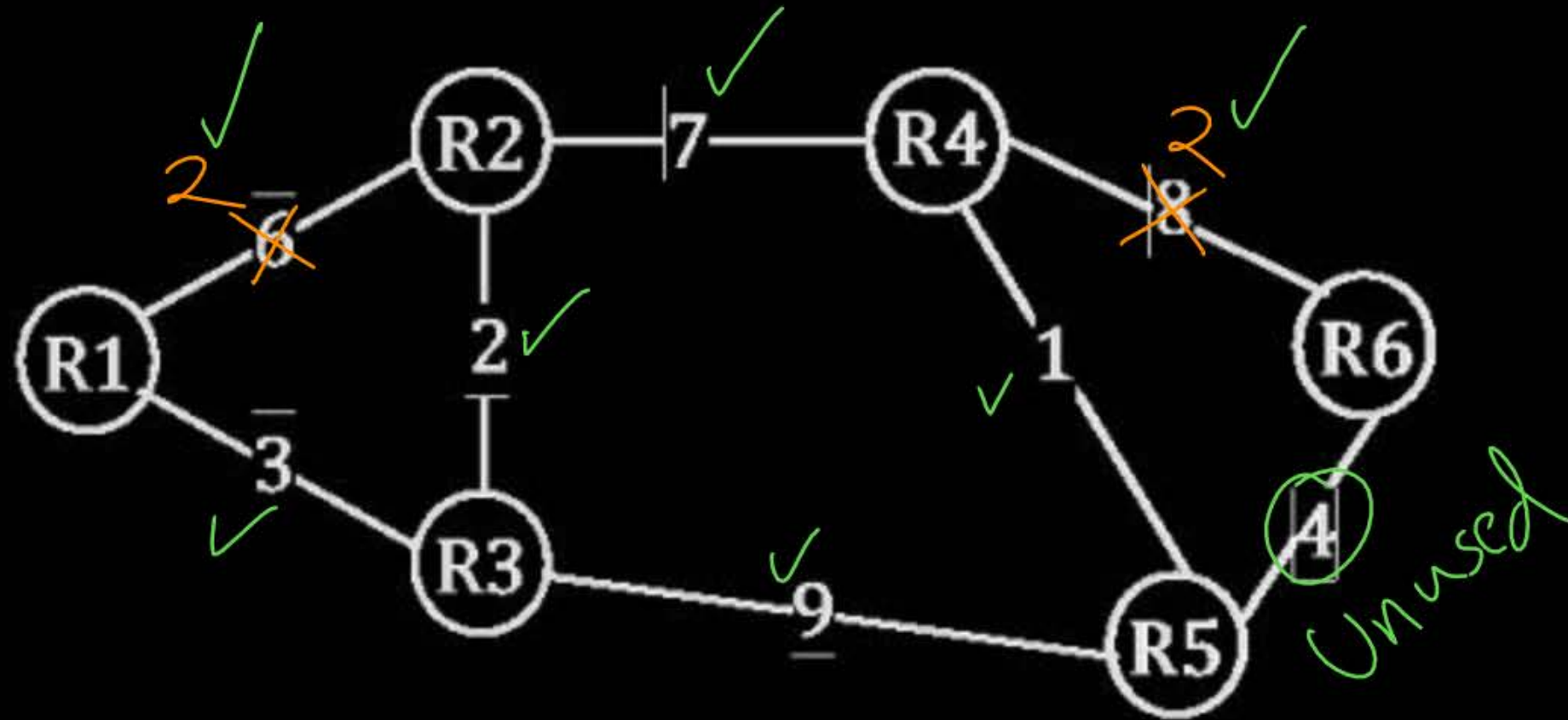
(A) 0

✓ (B) 1

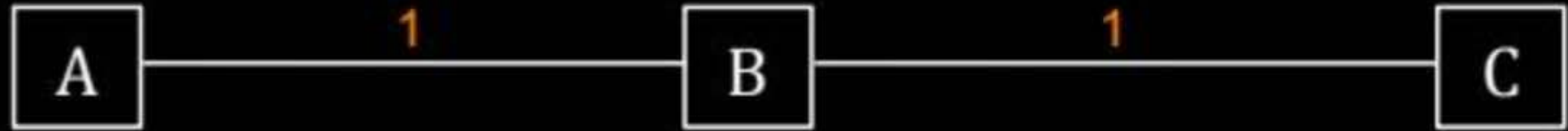
(C) 2

(D) 3

Ans: B



Intialization :-



Node	Cost	Via
A	0	A
B	1	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
B	1	B
C	0	C

First Round of Update :-



Node	Cost	Via
A	0	A
B	1	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
B	1	B
C	0	C



A	0
B	1



A	1
B	0
C	1




B	1
C	0

After First Round :-



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

Second Round of Update :-



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

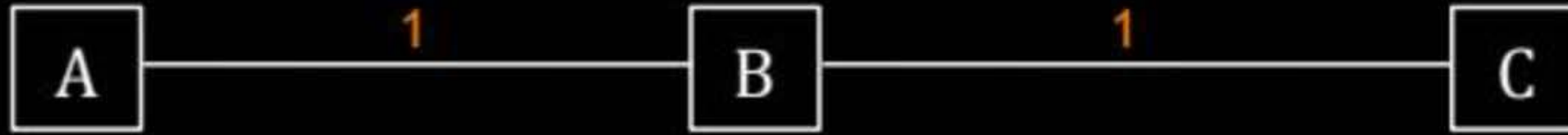


A	0
B	1
C	2



A	2
B	1
C	0

After Second Round :-



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

CASE I (First Round):-

After link down



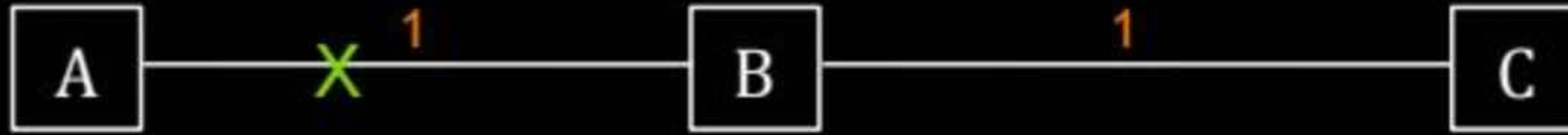
Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

A	∞
B	0
C	1

CASE I (Second Round):-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
<u>B</u>	<u>0</u>	B
<u>C</u>	<u>1</u>	<u>C</u>

Node	Cost	Via
A	∞	
<u>B</u>	<u>1</u>	<u>B</u>
<u>C</u>	<u>0</u>	<u>C</u>



A	∞
B	1
C	0



Topic : Distance Vector Routing



→ Problem : May suffer with "Count to infinity" problem
[Link to infinity problem, occur due to routing loop in DVR]

CASE II (First Round):- *After link down*



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C



A	∞
B	0
C	1



A	2
B	1
C	0

CASE II (Second Round):-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	3	C
B	0	B
C	1	C

Node	Cost	Via
A	∞	
B	1	B
C	0	C



A	3
B	0
C	1



A	∞
B	1
C	0

CASE II (Third Round):-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
<u>B</u>	<u>0</u>	<u>B</u>
<u>C</u>	<u>1</u>	<u>C</u>

Node	Cost	Via
A	4	B
<u>B</u>	<u>1</u>	<u>B</u>
<u>C</u>	<u>0</u>	<u>C</u>

A	∞
B	0
C	1

A	4
B	1
C	0

#Q. Count to infinity is a problem associated with :

[GATE-2005]

IIT-B

- ☒ (A) Link state routing protocol
- ☒ (B) Distance vector routing protocol
- ☒ (C) DNS while resolving host name
- ☒ (D) TCP for congestion control

Ans: B

#Q. Consider a network with three routers P, Q, R shown in the figure below. All the links have cost of unity.



The routers exchange distance vector routing information and have converged on the routing tables, after which the link Q–R fails. Assume that P and Q send out routing updates at random times, each at the same average rate. The probability of a routing loop formation (rounded off to one decimal place) between P and Q, leading to count-to-infinity problem, is _____.

[GATE-2022, NAT, 1-Mark]

IIT-KGP, H.W



Topic : Distance Vector Routing



→ Problem : May suffer with “Count to infinity” problem
[Link to infinity problem, occur due to routing loop in DVR]

Solution 1 : Advertise routing path in ‘Distance Vector’
[Send ‘Next Hop’ (via) with distance vector]



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C



Topic : Distance Vector Routing



→ **Problem** : May suffer with “**Count to infinity**” problem
[**Link to infinity** problem, occur due to routing loop in DVR]

Solution 2 : “**Split horizon hack**”
[Do not advertise the distance to a neighbor,
if the neighbor is the Next Hop for corresponding entry]



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	1	A
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C





Topic : Distance Vector Routing



→ **Problem** : May suffer with “**Count to infinity**” problem
[**Link to infinity** problem, occur due to routing loop in DVR]

Solution 3 : “**Poison reverse**”

[**Advertise infinite distance** to a neighbor,
if the **neighbor is the Next Hop** for corresponding entry]



Node	Cost	Via
A	0	A
B	1	B
C	2	B

Node	Cost	Via
A	<u>1</u>	<u>A</u>
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C

A	0
B	∞
C	∞

A	∞
B	0
C	1

A	1
B	0
C	∞

A	∞
B	∞
C	0



CASE III :-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
B	0	B
C	1	C

Node	Cost	Via
A	2	B
B	1	B
C	0	C



A	∞
B	0
C	∞



A	∞
B	∞
C	0

CASE III :-



Node	Cost	Via
A	0	A
B	∞	
C	∞	

Node	Cost	Via
A	∞	
B	0	B
C	1	C

Node	Cost	Via
A	∞	
B	1	B
C	0	C



Topic : Distance Vector Routing



→ **Problem** : May suffer with “Count to infinity” problem
[Link to infinity problem, occur due to routing loop in DVR]

Solution 1 : Advertise routing path in ‘Distance Vector’
[Send ‘Next Hop’ (via) with distance vector]

Solution 2 : “Split horizon hack”
[Do not advertise the distance to a neighbor,
if the neighbor is the Next Hop for corresponding entry]

Solution 3 : “Poison reverse”
[Advertise infinite distance to a neighbor,
if the neighbor is the Next Hop for corresponding entry]

#Q. Two popular routing algorithms are Distance Vector (DV) and Link State (LS) routing. Which of the following are true?

(S1) Count to infinity is a problem only with DV and not LS routing

(S2) In LS, the shortest path algorithm is run only at one node

(S3) In DV, the shortest path algorithm is run only at one node

(S4) DV requires lesser number of network messages than LS

(A) S1, S2 and S4 only

(B) S1, S3 and S4 only

(C) S2 and S3 only

(D) S1 and S4 only

11SC
[GATE-2008]
HW

#Q. Consider the following three statements about link state and distance vector routing protocols, for a large network with 500 network nodes and 4000 links.

[S1] The computational overhead in link state protocols is higher than in distance vector protocols.

[S2] A distance vector protocol (with split horizon) avoids persistent routing loops, but not a link state protocol.

[S3] After a topology change, a link state protocol will converge faster than a distance vector protocol.

Which one of the following is correct about S1, S2, and S3 ?

[GATE-2014, Set-1, 1-Mark]

- (A) S1, S2, and S3 are all true
- (B) S1, S2, and S3 are all false
- (C) S1 and S2 are true but S3 is false
- (D) S1 and S3 are true, but S2 is false

IT-KGP, H.W.

Statement for Linked Answer Questions.

Consider a network with five nodes, N1 to N5, as shown below. The network uses a Distance Vector Routing protocol. Once the routes have stabilized, the distance vectors at different nodes are as following.

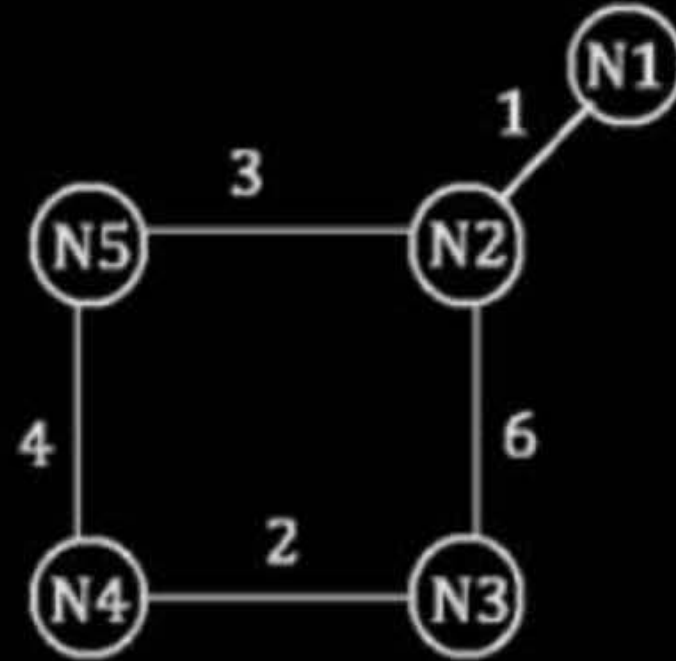
N1 : (0, 1, 7, 8, 4)

N2 : (1, 0, 6, 7, 3)

N3 : (7, 6, 0, 2, 6)

N4 : (8, 7, 2, 0, 4)

N5 : (4, 3, 6, 4, 0)



Each distance vector is the distance of the best known path at that instance to nodes, N1 to N5, where the distance to itself is 0. Also, all links are symmetric and the cost is identical in both directions. In each round, all nodes exchange their distance vectors with their respective neighbors. Then all nodes update their distance vectors. In between two rounds, any change in cost of a link will cause the two incident nodes to change only that entry in their distance vectors.

#Q. The cost of link N2-N3 reduces to 2 (in both directions). After the next round of updates, what will be the new distance vector at node, N3?

[GATE-2011]

IIT-M, H.W.

- (A) (3, 2, 0, 2, 5)
- (B) (3, 2, 0, 2, 6)
- (C) (7, 2, 0, 2, 5)
- (D) (7, 2, 0, 2, 6)



2 mins Summary



Topic

Distance Vector Routing



THANK - YOU

